Assignment 3

Trustworthy Machine Learning

Saarland University – Summer Semester 2025

Team 49/3
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1 Abstract

In this assignment, we implement methods to train a robust classifier with the highest possible accuracy for both clean and adversarial data points. In this report, we discuss potential solutions, our implementations, and several experiments, along with the results and insights we extracted from them.

2 Methodology

We implemented 2 different modeling approaches. They are listed below:

1. FSGM Adversarial Training
2. FAST

2.1 FSGM Adversarial Training.

We began training our robust classifier by training a simple resnet model using the FSGM Adversarial Training approach.

In our training we used a resnet-18 model taking epsilon as 8/255 or 0.31. We generated FSGM perturbations using the 1

$$x_{adv} = x + \epsilon * sign(grad) \tag{1}$$

We calculated scores for FGSM adversarial accuracy, PGD adversarial accuracy $\alpha = 0.007$ and clean accuracy scores.

The scores are given in Table 1 and Figure 1.

Evaluation	**Accuracy**
Final Clean Accuracy	0.6065
FGSM ($\epsilon = 0.03$) Adversarial Accuracy	0.3924
PGD ($\epsilon = 0.03$, iter=5) Adversarial Accuracy	0.3767

Table 1: Clean and Adversarial Accuracies for Single-Step FGSM Adversarial Training

2.2 FAST

Another method we implemented is FAST. Here, we used an epsilon and aplha values of 8/255 and 2/255 respectively. The accuracy values are given in table 2

Preprint. Under review.

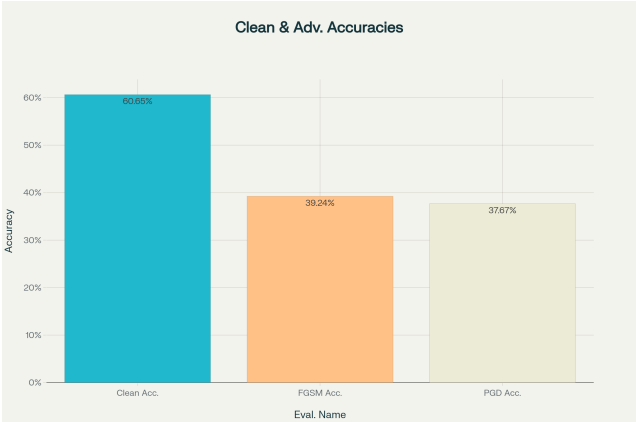

Figure 1: Accuracies on different sets for Single-Step FGSM Adversarial Training

Evaluation	**Accuracy**
Final Clean Accuracy	0.6092
FGSM ($\epsilon = 0.03$) Adversarial Accuracy	0.2783
PGD ($\epsilon = 0.03$, iter=5) Adversarial Accuracy	0.2782

Table 2: Clean and Adversarial Accuracies for Mixed Adversarial Training

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We show the clean accuracy for several epochs in 2

3 Directory Structure

The implementation has a folder called **contrastive** where all the files for contrastive learning are present:

1. fgsm.py : Implements single step FGSM.
2. hybrid.py: Implements FAST solution, mixed adversarial training.
3. shot.py: Our solution that gave best results logs: logs for different trainings.

We have training logs that show accuraies for various datasets.

The repository can be accessed here: `https://github.com/Cipher-unhsiV/TML25_A3_03/`

Acknowledgements

For Assignment lectures and paper [1], the reading from book [2] is referenced. For the implementation exercise, python3, pytorch [3] is used.

For making tables, `https://www.tablesgenerator.com/` is used.

References

[1] Jan Dubiński, Stanisław Pawlak, Franziska Boenisch, Tomasz Trzciński, and Adam Dziedzic. Bucks for buckets (b4b): Active defenses against stealing encoders, 2023.

[2] Daniel Jurafsky and James H. Martin. *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*. 3rd edition, 2024. Online manuscript released August 20, 2024; `https://web.stanford.edu/~jurafsky/slp3/`.

[3] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An imperative style, high-performance deep learning library. *Advances in neural information processing systems*, 32, 2019.

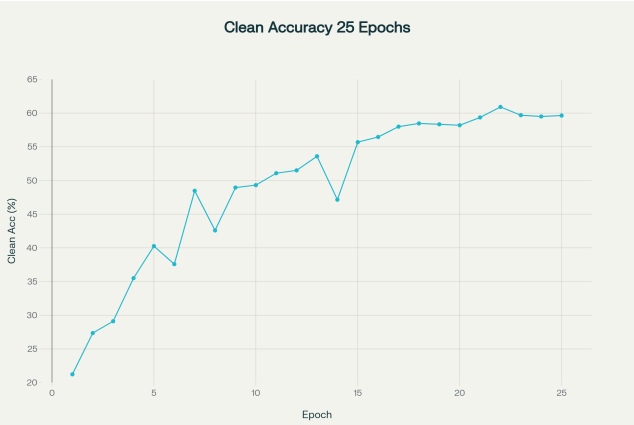

Figure 2: Clean Accuracies over different epochs for Mixed Adversarial Training